O-C-110C
October 18, 1984
SUPERSEDING
O-C-110B
May 14, 1974

FEDERAL SPECIFICATION

CALCIUM HYDROXIDE, TECHNICAL

This specification was approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers one technical grade of calcium hydroxide which is known commercially as hydrated lime.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

NN-P-71	-	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
UU-S-48	-	Sacks, Shipping, Paper
PPP-D-729	-	Drums, Shipping and Storage, Steel, 55-Gallon (208 Liters)

Federal Standards:

Fed. Std. No. 123	-	Marking for Shipment (Civil Agencies)
Fed. Std. No. 595	-	Colors

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.

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(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Washington, DC; Philadelphia, PA; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American Society for Testing and Materials (ASTM) Standards:

C 25 - Chemical Analysis of Limestone, Quicklime, and Hydrated Lime
C 50 - Sampling, Inspection, Packing, and Marking of Lime and Limestone Products
C 110 - Physical Testing of Quicklime, Hydrated Lime, and Limestone
D 1193 - Reagent Water
D 3951 - Commercial Packaging
E 29 - Indicating Which Places of Figures are to be Considered Significant in Specified Limiting Values

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Appearance. Calcium hydroxide shall be white, dry, finely powdered, and free from lumps or any foreign material that might interfere with the operation of dry-feed equipment, when tested as specified in 4.2.4.1.

3.2 Calcium hydroxide content. Calcium hydroxide shall contain no less than 90.0 percent by weight calcium hydroxide when tested as specified in 4.2.4.2.

3.3 Particle size characteristics. No less than 99.5 percent by weight of the calcium hydroxide shall pass through a 600-micrometer sieve and no less than 98.0 percent by weight shall pass through a 75-micrometer sieve when tested as specified in 4.2.4.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the calcium hydroxide produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling. See 6.4 for sampling and testing precautions.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For calcium hydroxide test. Sampling for test shall be conducted in accordance with ASTM C 50 using the procedure specified for sampling powdered materials in packages.

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4.2.2.3 For container leakage test. Sampling shall be conducted in accordance with MIL-STD-105, inspection level S-2.

4.2.3 Inspection procedure. See 6.4 for sampling and testing precautions.

4.2.3.1 For examination of packaging. The sample unit shall be one filled shipping container ready for shipment. Sample containers and the packaging thereof shall be examined for the following defects using inspection level S-2 and an AQL of 1.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing or illegible
(f) Unitization not as specified

4.2.3.2 For calcium hydroxide test. Each sample specimen collected and prepared as specified in ASTM C 50 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.3.3 For container leakage test. The sample unit shall be one container. The sample containers selected in 4.2.2.3 shall be tested as specified in 4.2.5 using an AQL of 1.5 percent defective.

4.2.3.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29.

4.2.4 Tests. See 6.4 for sampling and testing precautions. Water in accordance with ASTM D 1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Appearance. Visually examine the specimen for form, color, lumps, and foreign material.

4.2.4.2 Calcium hydroxide content. Determine the percent by weight calcium hydroxide in the specimen in accordance with ASTM C 25, using the method for Available Lime Index, Procedure for Hydrate.

4.2.4.3 Particle size characteristics. Determine the percent by weight specimen which passes through a 600-micrometer sieve and through a 75-micrometer sieve in accordance with ASTM C 110, using the method for Residue of Hydrated Lime.

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4.2.5 Container leakage test. Place the container on its side and rotate it about its longitudinal axis for three complete revolutions. Examine the container for any evidence of leakage.

5. PACKAGING

5.1 Unit packing. Calcium hydroxide shall be unit packed level A, B or commercial as specified (see 6.2).

5.1.1 Level A, 25-kilogram quantity. A quantity of 25 (+1/4 or -0) kilograms (kg) of calcium hydroxide shall be unit packed in a removable head steel drum conforming to Rule 40 of the Uniform Freight Classification. The exterior finish of the drum shall be green No. 24064 of Fed. Std. No. 595, applied in accordance with the exterior painting requirements for drums of PPP-D-729. The drum shall be closed tightly in accordance with the drum manufacturer's instructions. There shall be no evidence of sifting-out of contents when the unit pack is tested as specified in 4.2.5.

5.1.2 Level B, 25-kg quantity. A quantity of 25 (+1/4 or -0) kg of calcium hydroxide shall be unit packed level B in a multiwall paper shipping sack conforming to type optional, style B, construction number 8 or 8x, moisture barrier MB grade 2 of UU-S-48.

5.1.3 Commercial. A quantity of 25 (+1/4 or -0) kg of calcium hydroxide shall be unit packed commercially in accordance with ASTM D 3951.

5.2 Packing. Calcium hydroxide, unit packed as specified in 5.1, shall require no further protection for shipment other than unitization.

5.2.1 Unitization. Uniform quantities of containers of calcium hydroxide, packed level A or B, shall be palletized in accordance with MIL-STD-147, using softwood pallets conforming to type IV of NN-P-71, using load type III for drums, and load type XV for sacks. Commercial packs shall be unitized in a manner to assure carrier acceptance, safe delivery to destination, and stable stackability in storage for a minimum duration of 6 months.

5.3 Marking. Each container shall be marked to show the date of manufacture and batch or lot number of the calcium hydroxide. Commercial containers shall be marked in accordance with ASTM D 3951, and in addition with the weight and cube.

5.3.1 Civil agencies. Level A and B containers and pallet loads shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. Level A and B containers and pallet loads shall be marked in accordance with MIL-STD-129.

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5.3.3 Precautionary marking. Each container and pallet load shall be marked as follows:

WARNING!

Causes IRRITATION to skin, eyes and respiratory and digestive track.

FIRST AID: Remove contaminated clothing and shoes. Flush affected area with plenty of water. If in eyes, hold eyelids open and flush with water for at least 10 minutes. Call a physician.

If inhaled, move to fresh air and irrigate nose and throat. Call a physician.

If swallowed and victim is conscious, have victim drink water or milk. DO NOT INDUCE VOMITING.

6. NOTES

6.1 Intended use. Calcium hydroxide covered by this specification is intended for use in the treatment of sewage, industrial water, and potable water, as well as for general use. It is not intended for structural use such as in the manufacture of mortar or plaster.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification, and
(b) Level of unit packing required (see 5.1).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Sampling and testing precautions. This specification covers inspection of chemical material which is potentially dangerous to personnel. All applicable safety rules, regulations and procedures must be followed in the sampling, testing and handling of this material.

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MILITARY INTERESTS:

Custodians:

Army - EA
Air Force - 68

Review activities:

Army - MD, ME
DLA - GS

User activity:

Navy - MC

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-7FCE
NBS
VA-OSS

Preparing activity:

Army - EA

Project No. 6810-B463

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
O-C-110C	CALCIUM HYDROXIDE, TECHNICAL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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